INCH-POUND

MIL-DTL-32505C (MR)
10 November 2024
SUPERSEDING
MIL-DTL-32505B (MR)
15 January 2021

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM, ALLOYS 7017 and 7020, WELDABLE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wrought aluminum alloys AA7017 (class I) and AA7020 (class II) for welded and non-welded applications in nominal thicknesses from 0.500 to 4.000 inch, inclusive (see 6.2). Class I (AA7017) material is a direct replacement for MIL-DTL-46063H w/Amendment 2 material, namely AA7039 for new design and if specified (see 6.2), for old design or repair/replacement.

1.2 Weldability. The material classes covered by this specification have been demonstrated to be weldable to themselves and to other weldable alloys (see 6.3 and 6.4).

1.3 Classification. The wrought aluminum armor should be of the following classes, as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 7017 aluminum alloy. The applicable gauge range for class I is 0.500 - 4.000 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 7020 aluminum alloy. The applicable gauge range for class II is 0.500 - 4.000 inches.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TEST OPERATING PROCEDURES

TOP 2-2-711	-	Proofing Projectile Impact Testing of Armor Weldments

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products
Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/).

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ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

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3.2 Chemical composition. The chemical composition of the plates shall be within the limits shown in table I. The limits specified in table I were listed from the Aluminum Association Teal Sheets, International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, weight percent. 1/

ELEMENTS	SYMBOL	Class I 7017 ALLOY 2/	Class II 7020 ALLOY 2/
Silicon	Si	0.35	0.35
Iron	Fe	0.45	0.40
Copper	Cu	0.20	0.20
Manganese	Mn	0.05 - 0.50	0.05 - 0.50
Magnesium	Mg	2.0 - 3.0	1.0 - 1.4
Chromium	Cr	0.35	0.10 - 0.35
Nickel	Ni	0.10	N/A
Zinc	Zn	4.0 - 5.2	4.0 - 5.0
Titanium	Ti	0.15	N/A
Zirconium	Zr	0.10 - 0.25	0.08 - 0.20
Other, min	(Mn + Cr)	0.15	N/A
Other	(Zr + Ti)	N/A	0.08 - 0.25
Other, max. Each	---	0.05	0.05
Other, max. Total 3/	---	0.15	0.15
Aluminum	Al	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, except for [(Mn + Cr)] these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements equaling 0.010 percent or more for each, shall be expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal transverse (LT) direction shall meet the minimum mechanical properties listed in Table II. If mechanical property requirements differ from those contained in Table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE II. Minimum mechanical properties. 1/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class I	Class II	Class I	Class II	Class I	Class II
0.500 to 1.500, incl.	63.0	52.2	53.0	45.0	9	10
1.501 to 4.000, incl.	60.0	49.3	50.0	42.0	8	8

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.

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3.4 Weldability. When the vehicle requires weldments, the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum" for all armor to armor and armor to non-armor applications. This may also require Proofing Projectile Impact Testing (PPIT) for armor to armor weldments in accordance with procedures in TOP 2-2-711, "Proofing Projectile Impact Testing of Armor Weldments" (revision A and above) depending on the weld classification in accordance with MIL-STD-3057. PPIT procedures and associated striking velocities for established welded class I and class II plate thicknesses are listed in TOP 2-2-711. For any newly introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing following TOP 2-2-711. Since ordered thicknesses of the alloys referenced by this specification may not be listed in TOP 2-2-711, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5.1 Resistance to spalling. Class II (AA7020) material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or purchase order (see 6.2). Spall that arises as part of the V50 ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-32505 (see A.4.2).

3.6 Stress Corrosion Cracking Resistance. Unless otherwise specified in the contract or purchase order (see 6.2) plate material 0.750 inch and thicker in the short transverse direction shall be resistant to stress corrosion cracking. A minimum of 5 of the 9 specimens tested shall show no evidence of cracking at the end of 96 hours when tested as specified in 4.7.4.

3.7 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in Table III.

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TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.500	1.000		0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000		0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.000	4.000		0.090	0.090	0.105	0.110	0.125	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.9, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor, a properly completed Aluminum Armor Test Data Form (See Figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR										
SPECIFICATION: MIL-DTL-32505			**REVISION: C**			**AMENDMENT:**				
FIRING RECORD:		**DATE:**		**TEST CONTRACT NO:**						
	PLATE MANUFACTURER / PRODUCER			**PRIME CONTRACTOR**						
Name:										
Address:										
POC:										
Phone No:										
Email:										
TEST ITEM IDENTIFICATION:				Ordered Thickness:						
Lot No._____ **Plate No.**_____			**Alloy:**_____ **Temper:** _____			**Class:** _____ **Type:** _____				
PURPOSE:___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing										
SAMPLE: ____ Primary ____ Retest (Firing Record No. of Failed Sample _____)										
CHEMISTRY ANALYSIS:	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**	
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:										
UTS (ksi):			0.2% YS (ksi):			Elongation (%):				
WELDABILITY:	Thickness		Required Velocity		Actual Velocity		Pass/Fail			
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES ____ NO ____**										
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**					
Test:										
Projectile:										
	NOTES:									
SPALL:	Spall Size:		Two Projectile Calibers:		Pass / Fail:					
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT OFFICIAL ONLY										
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-32505										
Government Representative:		Date:	Supplier Representative:		Date:					

FIGURE 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all production and ballistic plates of the same alloy and from the same Direct Chill (DC) Cast, of the same source of molten metal representing the ingots poured, and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the DC ingot cast.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One (1) tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two (2) plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

4.5.1.4 Stress Corrosion Cracking. Nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.4.

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4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with Table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.3.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.4 Stress Corrosion Cracking. Nine test specimens shall be removed from one plate from each lot for testing in accordance with 4.7.4.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.9) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.8 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

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4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of any dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal transverse (LT) direction. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with Table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with Table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL (P) for that plate.

4.7.3.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	TABLE	CLASS I (7017)	CLASS II (7020)
0.500 - 0.749	Cal. .30 AP M2	30	A-I	LISTED	LISTED
0.750 - 0.950	Cal. .50 FSP	0	A-II	LISTED	LISTED
0.951 - 1.500	20-mm FSP	0	A-III	LISTED	LISTED
0.750 - 1.500	Cal. .30 AP M2	0	A-IV	LISTED	LISTED
1.501 - 2.500	Cal. .50 AP M2	0	A-V	LISTED	LISTED
2.501 - 4.000	14.5-mm BS41	0	A-VI	LISTED	LISTED

4.7.3.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

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4.7.4 Stress corrosion. Short transverse stress corrosion test specimens shall be prepared and tested in accordance with procedure outlined in ASTM G38 and ASTM G47. The specimens shall be stressed to 35 ksi. Nine specimens shall be tested per sample per lot. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense submit thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.4. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.3 and 4.7.4.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing that the manufacturer agrees in writing to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat and tactical vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Class II (AA7020) has better inherent resistance to spall.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3).
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties and ballistic requirements, if required (see 3.3).
(e) Size of spall that is considered a failure for AA7020 (see 3.5.1 and A.4.2).
(f) Dimension and tolerance requirements if different (see 3.8 and 3.8.1).
(g) If markings are different and when impression stamping is permissible (see 3.9).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If stress corrosion cracking resistance is not required or is different (see 3.6).
(j) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(k) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(l) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(m) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1).
(n) Rejection and retest requirement, if different (see 4.8).
(o) If reduced testing is allowed (see 4.9).
(p) Packaging requirements (see 5.1).
(q) Striking velocities for those thicknesses not covered (see A.3.6.1.1).

6.3 Weldability. During initial qualification of class I, AA7017 and class II, AA7020 alloys as armor plate, the Army Research Lab confirmed that class I, AA7017 and class II, AA7020 alloys can be welded and demonstrate good shock resistance when prepared and tested in accordance with MIL-STD-3057 and TOP 2-2-711. If these alloys in thicknesses required for manufacturing are not currently listed in TOP 2-2-711, the fabricator shall follow the procedures of TOP 2-2-711 to generate the required striking velocities before the Proofing Projectile Impact Testing (PPIT) is performed. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with MIL-STD-3057 and TOP 2-2-711. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 7000 and 5000 series alloys using a 5000 series filler metal. It is not weldable to 2000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated in accordance with MIL-STD-3057 and TOP 2-2-711 prior to acceptance by the procuring activity.

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6.4 Weld wire. Butt joints were successfully welded with AA5556 weld wire for class I (AA7017) when tested with the required proofing projectile per MIL-STD-3057 (see 1.2). For class II (AA7020) flat weldments (butt joints) AA5087 and AA5556 weld wires were successfully used in welding the plates for proof testing (see 1.2).

6.5 Density. The density of the specified alloys listed in this specification are as follows:

AA7017 = 0.100 lb/in^3 (2.76 g/cm^3)

AA7020 = 0.100 lb/in^3 (2.78 g/cm^3).

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.7 Subject term (key word) listing.

Ballistic testing	Caliber .50 FSP	20-mm FSP
Caliber .30 AP M2	M1114 HMMWV	Stress corrosion
Caliber .50 AP M2	Military vehicles	14.5-mm API

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ARMOR PLATE, ALUMINUM, ALLOY 7017 and 7020, WELDABLE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA7017 and AA7020 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any fair impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052-H36 aluminum alloy or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches (+ 1/2 inch) behind and parallel to the test plates or other ballistic sample.

A.3.6 Spalling. The detachment or delamination of a layer of material in the area surrounding the location of impact, which may occur on either the front or rear surfaces of the armor. Ejected spall may impact the witness plate and cause a CP even when the projectile has not exited the plate.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VI.

A.4.2 Resistance to spalling. Class II (AA7020) material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or purchase order (see 6.2). Spall that arises as part of the V50 ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-32505.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity

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is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with

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the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.415	1297	1255	0.560	1635	1535	0.705	1915	1772
0.420	1310	1266	0.565	1646	1544	0.710	1923	1780
0.425	1323	1276	0.570	1656	1553	0.715	1932	1787
0.430	1336	1287	0.575	1666	1562	0.720	1941	1795
0.435	1349	1297	0.580	1676	1570	0.725	1950	1802
0.440	1361	1308	0.585	1687	1579	0.730	1959	1810
0.445	1374	1318	0.590	1697	1587	0.735	1967	1817
0.450	1386	1328	0.595	1707	1596	0.740	1976	1824
0.455	1398	1338	0.600	1717	1604	0.745	1985	1832
0.460	1411	1348	0.605	1727	1613	0.749 [2/]	1993	1839
0.465	1423	1358	0.610	1737	1621	0.755	2002	1847
0.470	1435	1368	0.615	1746	1629	0.760	2010	1854
0.475	1446	1378	0.620	1756	1637	0.765	2019	1861
0.480	1458	1388	0.625	1766	1646	0.770	2027	1868
0.485	1470	1397	0.630	1775	1654	0.775	2036	1876
0.490	1481	1407	0.635	1785	1662	0.780	2044	1883
0.495	1493	1417	0.640	1795	1670	0.785	2052	1890
0.500 [1/]	1504	1426	0.645	1804	1678	0.790	2061	1897
0.505	1516	1435	0.650	1814	1686	0.795	2069	1904
0.510	1527	1445	0.655	1823	1694	0.800	2077	1911
0.515	1538	1454	0.660	1832	1702	0.805	2086	1918
0.520	1549	1463	0.665	1842	1710	0.810	2094	1925
0.525	1560	1473	0.670	1851	1718	0.815	2102	1932
0.530	1571	1482	0.675	1860	1726	0.820	2110	1939
0.535	1582	1491	0.680	1869	1733	0.825	2118	1946
0.540	1593	1500	0.685	1878	1741	0.830	2126	1953
0.545	1603	1509	0.690	1888	1749	0.835	2134	1960
0.550	1614	1518	0.695	1897	1757	---	---	---
0.555	1625	1527	0.700	1906	1764	---	---	---

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.665	1599	1565	0.790	2117	1998	0.915	2803	2549
0.670	1617	1581	0.795	2141	2017	0.920	2835	2574
0.675	1635	1596	0.800	2165	2037	0.925	2867	2599
0.680	1653	1612	0.805	2189	2057	0.930	2899	2625
0.685	1672	1628	0.810	2214	2077	0.935	2932	2650
0.690	1691	1644	0.815	2239	2097	0.940	2965	2676
0.695	1710	1660	0.820	2265	2118	0.945	2999	2703
0.700	1729	1676	0.825	2290	2139	0.950 [2/]	3033	2729
0.705	1749	1692	0.830	2316	2160	0.955	3067	2756
0.710	1769	1709	0.835	2342	2181	0.960	3102	2783
0.715	1789	1726	0.840	2369	2202	0.965	3137	2810
0.720	1809	1743	0.845	2395	2224	0.970	3172	2838
0.725	1829	1760	0.850	2422	2246	0.975	3208	2865
0.730	1850	1777	0.855	2450	2268	0.980	3244	2894
0.735	1871	1794	0.860	2477	2290	0.985	3281	2922
0.740	1892	1812	0.865	2505	2312	0.990	3318	2951
0.745	1913	1830	0.870	2534	2335	0.995	3355	2979
0.750 [1/]	1935	1848	0.875	2562	2358	1.000	3393	3009
0.755	1957	1866	0.880	2591	2381	1.005	3432	3038
0.760	1979	1884	0.885	2621	2404	1.010	3470	3068
0.765	2001	1902	0.890	2650	2428	1.015	3510	3098
0.770	2024	1921	0.895	2680	2451	1.020	3549	3128
0.775	2047	1940	0.900	2710	2476	1.025	3589	3159
0.780	2070	1959	0.905	2741	2500	1.030	3630	3190
0.785	2093	1978	0.910	2772	2524	1.035	3671	3221

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.865	1168	1162	1.035	1462	1444	1.205	1831	1793
0.870	1176	1170	1.040	1472	1453	1.210	1843	1805
0.875	1184	1177	1.045	1482	1462	1.215	1855	1816
0.880	1191	1185	1.050	1492	1471	1.220	1868	1828
0.885	1199	1192	1.055	1502	1481	1.225	1880	1839
0.890	1207	1200	1.060	1512	1490	1.230	1893	1851
0.895	1215	1208	1.065	1522	1500	1.235	1905	1863
0.900	1223	1215	1.070	1532	1510	1.240	1918	1875
0.905	1231	1223	1.075	1542	1519	1.245	1931	1887
0.910	1240	1231	1.080	1552	1529	1.250	1943	1899
0.915	1248	1239	1.085	1562	1539	1.255	1956	1911
0.920	1256	1247	1.090	1573	1549	1.260	1969	1923
0.925	1264	1255	1.095	1583	1558	1.265	1982	1936
0.930	1273	1263	1.100	1594	1568	1.270	1995	1948
0.935	1281	1271	1.105	1604	1578	1.275	2009	1961
0.940	1290	1279	1.110	1615	1589	1.280	2022	1973
0.945	1298	1287	1.115	1626	1599	1.285	2035	1986
0.951 1/	1309	1297	1.120	1636	1609	1.290	2049	1998
0.955	1316	1304	1.125	1647	1619	1.295	2063	2011
0.960	1324	1312	1.130	1658	1630	1.300	2076	2024
0.965	1333	1320	1.135	1669	1640	1.305	2090	2037
0.970	1342	1329	1.140	1680	1650	1.310	2104	2050
0.975	1351	1337	1.145	1691	1661	1.315	2118	2063
0.980	1360	1346	1.150	1703	1672	1.320	2132	2076
0.985	1369	1354	1.155	1714	1682	1.325	2146	2090
0.990	1378	1363	1.160	1725	1693	1.330	2160	2103
0.995	1387	1372	1.165	1737	1704	1.335	2175	2117
1.000	1396	1381	1.170	1748	1715	1.340	2189	2130
1.005	1406	1389	1.175	1760	1726	1.345	2204	2144
1.010	1415	1398	1.180	1772	1737	1.350	2218	2157
1.015	1424	1407	1.185	1783	1748	1.355	2233	2171
1.020	1434	1416	1.190	1795	1759	1.360	2248	2185
1.025	1443	1425	1.195	1807	1770	1.365	2263	2199
1.030	1453	1434	1.200	1819	1782	1.370	2278	2213

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.375	2293	2227	1.455	2549	2467	1.535	2833	2732
1.380	2308	2242	1.460	2565	2482	1.540	2852	2749
1.385	2323	2256	1.465	2582	2498	1.545	2871	2767
1.390	2339	2270	1.470	2600	2514	1.550	2890	2784
1.395	2354	2285	1.475	2617	2530	1.555	2909	2802
1.400	2370	2300	1.480	2634	2547	1.560	2928	2820
1.405	2385	2314	1.485	2652	2563	1.565	2948	2838
1.410	2401	2329	1.490	2669	2579	1.570	2967	2856
1.415	2417	2344	1.495	2687	2596	1.575	2987	2875
1.420	2433	2359	1.500 [2/]	2705	2612	1.580	3007	2893
1.425	2449	2374	1.505	2723	2629	1.585	3027	2911
1.430	2466	2389	1.510	2741	2646	1.590	3047	2930
1.435	2482	2405	1.515	2759	2663	1.595	3067	2949
1.440	2498	2420	1.520	2777	2680	1.600	3087	2968
1.445	2515	2435	1.525	2796	2697	1.605	3108	2987
1.450	2532	2451	1.530	2814	2714	---	---	---

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.665	1647	1527	0.835	1903	1773	1.005	2129	1990
0.670	1655	1534	0.840	1910	1780	1.010	2135	1996
0.675	1663	1542	0.845	1917	1787	1.015	2141	2002
0.680	1671	1550	0.850	1924	1793	1.020	2148	2008
0.685	1679	1558	0.855	1931	1800	1.025	2154	2014
0.690	1687	1565	0.860	1938	1807	1.030	2160	2020
0.695	1695	1573	0.865	1945	1813	1.035	2166	2025
0.70	1703	1580	0.870	1952	1820	1.040	2172	2031
0.705	1711	1588	0.875	1959	1827	1.045	2179	2037
0.710	1718	1596	0.880	1965	1833	1.050	2185	2043
0.715	1726	1603	0.885	1972	1840	1.055	2191	2049
0.720	1734	1611	0.890	1979	1846	1.060	2197	2055
0.725	1742	1618	0.895	1986	1853	1.065	2203	2061
0.730	1749	1625	0.900	1993	1859	1.070	2209	2066
0.735	1757	1633	0.905	1999	1865	1.075	2215	2072
0.740	1765	1640	0.910	2006	1872	1.080	2221	2078
0.745	1772	1647	0.915	2013	1878	1.085	2227	2084
0.750 1/	1780	1655	0.920	2019	1885	1.090	2233	2090
0.755	1787	1662	0.925	2026	1891	1.095	2239	2095
0.760	1795	1669	0.930	2032	1897	1.100	2245	2101
0.765	1802	1676	0.935	2039	1904	1.105	2251	2107
0.770	1810	1683	0.940	2046	1910	1.110	2257	2112
0.775	1817	1690	0.945	2052	1916	1.115	2263	2118
0.780	1824	1697	0.950	2059	1922	1.120	2269	2124
0.785	1832	1704	0.955	2065	1929	1.125	2275	2129
0.790	1839	1711	0.960	2072	1935	1.130	2281	2135
0.795	1846	1718	0.965	2078	1941	1.135	2286	2140
0.800	1853	1725	0.970	2084	1947	1.140	2292	2146
0.805	1861	1732	0.975	2091	1953	1.145	2298	2152
0.810	1868	1739	0.980	2097	1959	1.150	2304	2157
0.815	1875	1746	0.985	2104	1965	1.155	2310	2163
0.820	1882	1753	0.990	2110	1972	1.160	2316	2168
0.825	1889	1760	0.995	2116	1978	1.165	2321	2174
0.830	1896	1767	1.000	2123	1984	1.170	2327	2179

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.175	2333	2185	1.320	2494	2338	1.465	2645	2482
1.180	2339	2190	1.325	2499	2343	1.470	2650	2487
1.185	2344	2196	1.330	2504	2349	1.475	2655	2492
1.190	2350	2201	1.335	2510	2354	1.480	2660	2497
1.195	2356	2207	1.340	2515	2359	1.485	2665	2502
1.200	2361	2212	1.345	2520	2364	1.490	2670	2506
1.205	2367	2217	1.350	2526	2369	1.495	2675	2511
1.210	2373	2223	1.355	2531	2374	1.500 [2/]	2680	2516
1.215	2378	2228	1.360	2536	2379	1.505	2685	2521
1.220	2384	2234	1.365	2541	2384	1.510	2690	2525
1.225	2389	2239	1.370	2547	2389	1.515	2695	2530
1.230	2395	2244	1.375	2552	2394	1.520	2700	2535
1.235	2401	2250	1.380	2557	2399	1.525	2705	2540
1.240	2406	2255	1.385	2562	2404	1.530	2710	2544
1.245	2412	2260	1.390	2568	2409	1.535	2715	2549
1.250	2417	2265	1.395	2573	2414	1.540	2719	2554
1.255	2423	2271	1.400	2578	2419	1.545	2724	2558
1.260	2428	2276	1.405	2583	2424	1.550	2729	2563
1.265	2434	2281	1.410	2588	2429	1.555	2734	2568
1.270	2439	2287	1.415	2594	2434	1.560	2739	2572
1.275	2445	2292	1.420	2599	2439	1.565	2744	2577
1.280	2450	2297	1.425	2604	2443	1.570	2749	2582
1.285	2456	2302	1.430	2609	2448	1.575	2754	2586
1.290	2461	2307	1.435	2614	2453	1.580	2759	2591
1.295	2467	2313	1.440	2619	2458	1.585	2763	2596
1.300	2472	2318	1.455	2624	2463	1.590	2768	2600
1.305	2477	2323	1.445	2629	2468	1.595	2773	2605
1.310	2483	2328	1.450	2634	2473	1.600	2778	2609
1.315	2488	2333	1.460	2640	2478	1.605	2783	2614

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.395	1903	1772	1.570	2043	1916	1.745	2174	2050
1.400	1907	1776	1.575	2047	1920	1.750	2178	2054
1.405	1911	1780	1.580	2051	1924	1.755	2182	2057
1.410	1915	1785	1.585	2055	1928	1.760	2185	2061
1.415	1919	1789	1.590	2059	1932	1.765	2189	2065
1.420	1924	1793	1.595	2062	1936	1.770	2192	2068
1.425	1928	1797	1.600	2066	1940	1.775	2196	2072
1.430	1932	1801	1.605	2070	1943	1.780	2200	2076
1.435	1936	1806	1.610	2074	1947	1.785	2203	2079
1.440	1940	1810	1.615	2078	1951	1.790	2207	2083
1.445	1944	1814	1.620	2081	1955	1.795	2210	2087
1.450	1948	1818	1.625	2085	1959	1.800	2214	2090
1.455	1952	1822	1.630	2089	1963	1.805	2217	2094
1.460	1956	1827	1.635	2093	1967	1.810	2221	2098
1.465	1960	1831	1.640	2097	1971	1.815	2225	2101
1.470	1964	1835	1.645	2100	1974	1.820	2228	2105
1.475	1968	1839	1.650	2104	1978	1.825	2232	2108
1.480	1972	1843	1.655	2108	1982	1.830	2235	2112
1.485	1976	1847	1.660	2112	1986	1.835	2239	2116
1.490	1980	1851	1.665	2115	1990	1.840	2242	2119
1.495	1984	1855	1.670	2119	1994	1.845	2246	2123
1.501 1/	1989	1860	1.675	2123	1997	1.850	2249	2126
1.505	1992	1864	1.680	2126	2001	1.855	2253	2130
1.510	1996	1868	1.685	2130	2005	1.860	2256	2133
1.515	2000	1872	1.690	2134	2009	1.865	2260	2137
1.520	2004	1876	1.695	2138	2013	1.870	2263	2141
1.525	2008	1880	1.700	2141	2016	1.875	2267	2144
1.530	2012	1884	1.705	2145	2020	1.880	2270	2148
1.535	2016	1888	1.710	2149	2024	1.885	2274	2151
1.540	2020	1892	1.715	2152	2028	1.890	2277	2155
1.545	2024	1896	1.720	2156	2031	1.895	2281	2158
1.550	2028	1900	1.725	2160	2035	1.900	2284	2162
1.555	2031	1904	1.730	2163	2039	1.905	2288	2165
1.560	2035	1908	1.735	2167	2042	1.910	2291	2169
1.565	2039	1912	1.740	2171	2046	1.915	2295	2172

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.920	2298	2176	2.100	2419	2298	2.280	2534	2414
1.925	2301	2179	2.105	2422	2301	2.285	2537	2417
1.930	2305	2183	2.110	2425	2305	2.290	2540	2420
1.935	2308	2186	2.115	2428	2308	2.295	2543	2423
1.940	2312	2190	2.120	2432	2311	2.300	2546	2427
1.945	2315	2193	2.125	2435	2314	2.305	2549	2430
1.950	2319	2197	2.130	2438	2318	2.310	2552	2433
1.955	2322	2200	2.135	2441	2321	2.315	2555	2436
1.960	2325	2203	2.140	2445	2324	2.320	2558	2439
1.965	2329	2207	2.145	2448	2328	2.325	2562	2442
1.970	2332	2210	2.150	2451	2331	2.330	2565	2445
1.975	2336	2214	2.155	2454	2334	2.335	2568	2448
1.980	2339	2217	2.160	2458	2337	2.340	2571	2451
1.985	2342	2221	2.165	2461	2341	2.345	2574	2455
1.990	2346	2224	2.170	2464	2344	2.350	2577	2458
1.995	2349	2227	2.175	2467	2347	2.355	2580	2461
2.000	2352	2231	2.180	2470	2350	2.360	2583	2464
2.005	2356	2234	2.185	2474	2353	2.365	2586	2467
2.010	2359	2238	2.190	2477	2357	2.370	2589	2470
2.015	2362	2241	2.195	2480	2360	2.375	2592	2473
2.020	2366	2244	2.200	2483	2363	2.380	2595	2476
2.025	2369	2248	2.205	2486	2366	2.385	2598	2479
2.030	2372	2251	2.210	2490	2370	2.390	2601	2482
2.035	2376	2255	2.215	2493	2373	2.395	2604	2485
2.040	2379	2258	2.220	2496	2376	2.400	2607	2488
2.045	2382	2261	2.225	2499	2379	2.405	2610	2491
2.050	2386	2265	2.230	2502	2382	2.410	2613	2494
2.055	2389	2268	2.235	2505	2386	2.415	2616	2497
2.060	2392	2271	2.240	2508	2389	2.420	2619	2501
2.065	2396	2275	2.245	2512	2392	2.425	2622	2504
2.070	2399	2278	2.250	2515	2395	2.430	2626	2507
2.075	2402	2281	2.255	2518	2398	2.435	2629	2510
2.080	2406	2285	2.260	2521	2401	2.440	2632	2513
2.085	2409	2288	2.265	2524	2405	2.445	2635	2516
2.090	2412	2291	2.270	2527	2408	2.450	2638	2519
2.095	2415	2295	2.275	2530	2411	2.455	2641	2522

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
2.460	2644	2525	2.520	2679	2561	2.580	2714	2596
2.465	2646	2528	2.525	2682	2564	2.585	2717	2599
2.470	2649	2531	2.530	2685	2566	2.590	2720	2602
2.475	2652	2534	2.535	2688	2569	2.595	2723	2605
2.480	2655	2537	2.540	2691	2572	2.600	2726	2608
2.485	2658	2540	2.545	2694	2575	2.605	2729	2610
2.490	2661	2543	2.550	2697	2578	2.610	2732	2613
2.495	2664	2546	2.555	2700	2581	2.615	2735	2616
2.500 [2/]	2667	2549	2.560	2703	2584	2.620	2738	2619
2.505	2670	2552	2.565	2706	2587	2.625	2740	2622
2.510	2673	2555	2.570	2708	2590	---	---	---
2.515	2676	2558	2.575	2711	2593	---	---	---

[2/] Specification requirements end for this ordered thickness

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
2.375	2357	2221	2.550	2468	2333	2.725	2575	2440
2.380	2360	2224	2.555	2471	2336	2.730	2577	2443
2.385	2364	2228	2.560	2474	2339	2.735	2580	2446
2.390	2367	2231	2.565	2477	2343	2.740	2583	2449
2.395	2370	2234	2.570	2481	2346	2.745	2586	2452
2.400	2373	2237	2.575	2484	2349	2.750	2589	2455
2.405	2377	2241	2.580	2487	2352	2.755	2592	2458
2.410	2380	2244	2.585	2490	2355	2.760	2595	2461
2.415	2383	2247	2.590	2493	2358	2.765	2598	2464
2.420	2386	2250	2.595	2496	2361	2.770	2601	2467
2.425	2389	2254	2.600	2499	2364	2.775	2604	2470
2.430	2393	2257	2.605	2502	2367	2.780	2607	2473
2.435	2396	2260	2.610	2505	2370	2.785	2610	2476
2.440	2399	2263	2.615	2508	2374	2.790	2613	2479
2.445	2402	2267	2.620	2511	2377	2.795	2616	2482
2.450	2405	2270	2.625	2514	2380	2.800	2619	2485
2.455	2409	2273	2.630	2517	2383	2.805	2622	2488
2.460	2412	2276	2.635	2520	2386	2.810	2625	2490
2.465	2415	2279	2.640	2523	2389	2.815	2627	2493
2.470	2418	2283	2.645	2526	2392	2.820	2630	2496
2.475	2421	2286	2.650	2529	2395	2.825	2633	2499
2.480	2424	2289	2.655	2533	2398	2.830	2636	2502
2.485	2428	2292	2.660	2536	2401	2.835	2639	2505
2.490	2431	2295	2.665	2539	2404	2.840	2642	2508
2.495	2434	2299	2.670	2542	2407	2.845	2645	2511
2.501 [1/]	2438	2302	2.675	2545	2410	2.850	2648	2514
2.505	2440	2305	2.680	2548	2413	2.855	2651	2517
2.510	2443	2308	2.685	2551	2416	2.860	2654	2520
2.515	2446	2311	2.690	2554	2419	2.865	2656	2522
2.520	2450	2314	2.695	2557	2422	2.870	2659	2525
2.525	2453	2318	2.700	2560	2425	2.875	2662	2528
2.530	2456	2321	2.705	2563	2428	2.880	2665	2531
2.535	2459	2324	2.710	2566	2431	2.885	2668	2534
2.540	2462	2327	2.715	2569	2434	2.890	2671	2537
2.545	2465	2330	2.720	2572	2437	2.895	2674	2540

[1/] Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
2.900	2677	2543	3.075	2775	2641	3.250	2870	2736
2.905	2679	2545	3.080	2778	2644	3.255	2873	2739
2.910	2682	2548	3.085	2780	2647	3.260	2875	2741
2.915	2685	2551	3.090	2783	2649	3.265	2878	2744
2.920	2688	2554	3.095	2786	2652	3.270	2881	2747
2.925	2691	2557	3.100	2789	2655	3.275	2883	2749
2.930	2694	2560	3.105	2791	2658	3.280	2886	2752
2.935	2697	2563	3.110	2794	2660	3.285	2888	2755
2.940	2699	2565	3.115	2797	2663	3.290	2891	2757
2.945	2702	2568	3.120	2800	2666	3.295	2894	2760
2.950	2705	2571	3.125	2802	2669	3.300	2896	2763
2.955	2708	2574	3.130	2805	2671	3.305	2899	2765
2.960	2711	2577	3.135	2808	2674	3.310	2902	2768
2.965	2714	2580	3.140	2811	2677	3.315	2904	2770
2.970	2716	2582	3.145	2813	2679	3.320	2907	2773
2.975	2719	2585	3.150	2816	2682	3.325	2910	2776
2.980	2722	2588	3.155	2819	2685	3.330	2912	2778
2.985	2725	2591	3.160	2821	2688	3.335	2915	2781
2.990	2728	2594	3.165	2824	2690	3.340	2918	2784
2.995	2730	2597	3.170	2827	2693	3.345	2920	2786
3.000	2733	2599	3.175	2830	2696	3.350	2923	2789
3.005	2736	2602	3.180	2832	2698	3.355	2925	2791
3.010	2739	2605	3.185	2835	2701	3.360	2928	2794
3.015	2742	2608	3.190	2838	2704	3.365	2931	2797
3.020	2744	2611	3.195	2840	2707	3.370	2933	2799
3.025	2747	2613	3.200	2843	2709	3.375	2936	2802
3.030	2750	2616	3.205	2846	2712	3.380	2938	2804
3.035	2753	2619	3.210	2848	2715	3.385	2941	2807
3.040	2756	2622	3.215	2851	2717	3.390	2944	2810
3.045	2758	2624	3.220	2854	2720	3.395	2946	2812
3.050	2761	2627	3.225	2856	2723	3.400	2949	2815
3.055	2764	2630	3.230	2859	2725	3.405	2951	2817
3.060	2767	2633	3.235	2862	2728	3.410	2954	2820
3.065	2769	2636	3.240	2865	2731	3.415	2957	2823
3.070	2772	2638	3.245	2867	2733	3.420	2959	2825

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
3.425	2962	2828	3.605	3053	2919	3.785	3142	3008
3.430	2964	2830	3.610	3056	2922	3.790	3145	3010
3.435	2967	2833	3.615	3058	2924	3.795	3147	3013
3.440	2970	2836	3.620	3061	2927	3.800	3150	3015
3.445	2972	2838	3.625	3063	2929	3.805	3152	3017
3.450	2975	2841	3.630	3066	2932	3.810	3155	3020
3.455	2977	2843	3.635	3068	2934	3.815	3157	3022
3.460	2980	2846	3.640	3071	2937	3.820	3159	3025
3.465	2982	2848	3.645	3073	2939	3.825	3162	3027
3.470	2985	2851	3.650	3076	2942	3.830	3164	3029
3.475	2988	2853	3.655	3078	2944	3.835	3167	3032
3.480	2990	2856	3.660	3081	2946	3.840	3169	3034
3.485	2993	2859	3.665	3083	2949	3.845	3172	3037
3.490	2995	2861	3.670	3086	2951	3.850	3174	3039
3.495	2998	2864	3.675	3088	2954	3.855	3176	3041
3.500	3000	2866	3.680	3091	2956	3.860	3179	3044
3.505	3003	2869	3.685	3093	2959	3.865	3181	3046
3.510	3005	2871	3.690	3096	2961	3.870	3184	3049
3.515	3008	2874	3.695	3098	2964	3.875	3186	3051
3.520	3011	2876	3.700	3101	2966	3.880	3188	3053
3.525	3013	2879	3.705	3103	2969	3.885	3191	3056
3.530	3016	2881	3.710	3106	2971	3.890	3193	3058
3.535	3018	2884	3.715	3108	2974	3.895	3196	3061
3.540	3021	2886	3.720	3111	2976	3.900	3198	3063
3.545	3023	2889	3.725	3113	2978	3.905	3200	3065
3.550	3026	2892	3.730	3116	2981	3.910	3203	3068
3.555	3028	2894	3.735	3118	2983	3.915	3205	3070
3.560	3031	2897	3.740	3120	2986	3.920	3208	3072
3.565	3033	2899	3.745	3123	2988	3.925	3210	3075
3.570	3036	2902	3.750	3125	2991	3.930	3212	3077
3.575	3038	2904	3.755	3128	2993	3.935	3215	3080
3.580	3041	2907	3.760	3130	2996	3.940	3217	3082
3.585	3043	2909	3.765	3133	2998	3.945	3219	3084
3.590	3046	2912	3.770	3135	3000	3.950	3222	3087
3.595	3048	2914	3.775	3138	3003	3.955	3224	3089
3.600	3051	2917	3.780	3140	3005	3.960	3227	3091

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
3.965	3229	3094	4.020	3255	3120	4.075	3281	3145
3.970	3231	3096	4.025	3257	3122	4.080	3283	3148
3.975	3234	3098	4.030	3260	3124	4.085	3285	3150
3.980	3236	3101	4.035	3262	3127	4.090	3288	3152
3.985	3238	3103	4.040	3264	3129	4.095	3290	3154
3.990	3241	3106	4.045	3267	3131	4.100	3292	3157
3.995	3243	3108	4.050	3269	3134	4.105	3295	3159
4.000 2/	3246	3110	4.055	3271	3136	4.110	3297	3161
4.005	3248	3113	4.060	3274	3138	4.115	3299	3164
4.010	3250	3115	4.065	3276	3141	4.120	3302	3166
4.015	3253	3117	4.070	3278	3143	4.125	3304	3168

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Air Force - 11
Navy - SH

Preparing activity:
ARMY - MR

(Project 9535-2024-003)

Review activities:
Army - AT, TE
Navy - SH
DLA - CQ

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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